VIMAL JYOTHI ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to APJ Abdul Kalam Technological University)

B.TECH DEGREE VIII SEMESTER INTERNAL EXAM RETEST | 2023 BATCH

CET415: ENVIRONMENTAL IMPACT ASSESSMENT

SCHEME OF VALUATION / ANSWER KEY

Max. Marks: 60 **Duration:** 2 hours

PART A

Answer all questions. Each question carries 3 marks.

Q. No.	Answer / valuation points	CO	Marks
1	**1. Definition of drinking water quality parameter and correct standard** • Any three valid parameter-standard pairs: pH 6.5–8.5; TDS 500 mg/L acceptable; fluoride 1.0 mg/L acceptable; chloride 250 mg/L acceptable; nitrate 45 mg/L; turbidity 1 NTU acceptable. → 1 mark each	CO1	(3)
2	**2. Significance of public participation in EIA** • Identification of local environmental/social concerns → 1 mark • Opportunity for the public to express opinions and concerns → 1 mark • Transparency, better decisions, mitigation or conflict reduction → 1 mark	CO1	(3)
3	**3. Noise pollution: definition and sources** • Correct definition of noise pollution → 1 mark • Any two valid sources (traffic, aircraft, machinery, construction, loudspeakers, etc.) → 1 mark each	CO3	(3)
4	**4. Definitions: Impact Prediction and Socio-economic Impacts** • Impact prediction: estimating likely environmental changes caused by a project → 1.5 marks • Socio-economic impacts: changes in livelihood, employment, health, income or community life → 1.5 marks	CO3	(3)
5	**5. Water impacts of solid waste management project and mitigation** • First valid water impact with relevant mitigation → 1 mark • Second distinct water impact with relevant mitigation → 1 mark • Additional technical detail (liner, leachate treatment, drainage, monitoring, etc.) → 1 mark	CO5	(3)
6	**6. Impacts of a quarry mining project** • Any three distinct impacts, 1 mark each: dust/air pollution; noise/vibration; land degradation; habitat loss; erosion; drainage or groundwater impacts; safety risks	CO5	(3)

PART B

Answer one question from each OR group, as instructed in the question paper. Each selected question/group carries 14 marks.

Group 1: Q7 OR Q8

Question	Valuation points	CO	Marks
Q7(a)	**Q7(a). Need for EIA in developmental projects (9 marks)** • Definition/purpose of EIA → 1 mark • Early identification of impacts → 1 mark • Informed decision-making → 1 mark • Analysis of alternatives → 1 mark • Mitigation planning → 1 mark • Legal compliance → 1 mark • Public participation → 1 mark • Resource conservation / sustainable development → 1 mark • Monitoring and accountability → 1 mark	CO1	(9)
Q7(b)	**Q7(b). Limitations of EIA (5 marks)** • Any five valid limitations, 1 mark each: incomplete baseline data; prediction uncertainty; high cost/time; weak implementation; bias; limited public participation; cumulative impacts overlooked; long-term effects difficult to predict.	CO1	(5)

Question	Valuation points	CO	Marks
Q8	**Q8. Water (Prevention and Control of Pollution) Act, 1974 (14 marks)** • Objectives and introduction → 2 marks • CPCB functions → 2 marks • SPCB functions → 2 marks • Control of discharges / Section 24 → 2 marks • Consent provisions / Sections 25 and 26 → 2 marks • Inspection, sampling and laboratories → 2 marks • Enforcement, penalties and significance → 2 marks	CO1	(14)

SCHEME OF VALUATION / ANSWER KEY (CONTINUED)

Group 2: Q9 OR Q10

Question	Valuation points	CO	Marks
Q9	**Q9. Prediction, assessment, valuation and mitigation of air impacts (14 marks)** • Identification of air pollution sources → 2 marks • Baseline air-quality monitoring and meteorology → 2 marks • Emission estimation and impact prediction → 2 marks • Comparison with standards and exposure assessment → 2 marks • Valuation and significance assessment → 2 marks • Mitigation measures (dust suppression, emission controls, traffic management, etc.) → 3 marks • Monitoring and corrective action → 1 mark	CO3	(14)
Q10	**Q10. Role of air-quality modelling in impact assessment (14 marks)** • Definition and purpose → 2 marks • Input data: emissions, meteorology, terrain, background concentrations → 3 marks • Modelling procedure and predicted ground-level concentrations → 3 marks • Types of models and appropriate uses → 2 marks • Benefits for identifying hotspots and testing mitigation → 2 marks • Limitations, validation and monitoring → 2 marks	CO3	(14)

Group 3: Q11 OR Q12

Question	Valuation points	CO	Marks
Q11	**Q11. EIA case study of a highway-widening project in Kerala (14 marks)** • Project description and objectives → 2 marks • Baseline environmental conditions → 2 marks • Construction and operational impacts → 3 marks • Social impacts and land acquisition → 2 marks • Mitigation measures → 2 marks • Environmental Management Plan and monitoring → 2 marks • Public participation and conclusion → 1 mark	CO5	(14)
Q12	**Q12. Generic structure of EIA for an airport project (14 marks)** • Executive summary and project description → 2 marks • Regulatory framework and baseline data → 2 marks • Analysis of alternatives → 1 mark • Impact identification and prediction → 3 marks • Impact evaluation → 1 mark • Mitigation measures → 2 marks • Environmental Management Plan and monitoring → 1 mark • Risk/disaster management → 1 mark • Public consultation and conclusion → 1 mark	CO5	(14)

GENERAL EVALUATION GUIDELINES FOR EVALINK

- Accept correct paraphrases and equivalent technically valid explanations. Exact wording is not required.
- Award partial marks according to the listed criteria. Do not award marks solely because a keyword appears.
- Do not double-count repeated ideas. Accept equivalent examples and valid alternative case studies.
- Do not penalize grammar or spelling unless meaning becomes unclear.
- For Q11, accept any genuine highway-widening project in Kerala if the EIA components are explained correctly.
- For legal details in Q8, evaluate the legal principle accurately and account for amendments; verify section-specific claims.
- Keep OCR confidence separate from grading confidence. Flag illegible, ambiguous or low-confidence answers for human review.
- This is a proposed valuation scheme based on the supplied question paper, not an official college-approved scheme. Subject-expert review is recommended before live grading.